

PAN-AQI: A Panoramic Dataset and Framework for Urban Air Quality Estimation

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Abstract—Image-based air quality estimation offers a scalable alternative to sparse sensor-based monitoring. However, existing methods rely on narrow field-of-view imagery and overlook both the spatial structure of panoramic scenes and the ordinal nature of Air Quality Index (AQI) prediction. To address these limitations, we introduce PAN-AQI, a large-scale dataset comprising 33,982 panoramic 360° street-view images collected over 1000 km across the twin city of Hyderabad and Secunderabad, India, with co-located PM_{2.5}, PM₁₀, Temperature, and Humidity measurements. We further propose PANQIFormer, a multi-modal transformer framework that jointly models panoramic visual context and environmental metadata for AQI estimation. The proposed architecture combines spatial-zone reasoning, attention-based multimodal fusion, and ordinal-aware learning to perform both AQI category classification and continuous AQI regression. Experiments demonstrate consistent improvements over previous image-based AQI methods across both tasks, while ablation studies show that panoramic coverage, spatial reasoning, and multimodal fusion each contribute substantially to performance. Together, PAN-AQI and PANQIFormer establish panoramic sensing as a promising direction for scalable vision-based air-quality monitoring.

Index Terms—Air quality estimation, panoramic sensing, 360° imagery, vision transformers, multimodal learning, ordinal classification, environmental monitoring.

I. INTRODUCTION

Air pollution poses a significant global health concern, with PM_{2.5} and PM₁₀ being the primary pollutants responsible for the deterioration of ambient air quality in urban environments in India [1], [2]. The Air Quality Index (AQI) is a standardised metric ranging from 0 to 500 that communicates the level of air pollution to the public, where lower values indicate better air quality. The Central Pollution Control Board (CPCB), India, has defined six AQI categories: Good (0–50), Satisfactory (51–100), Moderate (101–200), Poor (201–300), Very Poor (301–400), and Severe (>400) [3]. India is among the most severely affected countries, consistently ranking in the top three for population-weighted PM_{2.5} burden [4], and the National Clean Air Programme has identified more than 100 non-attainment cities that persistently exceed national ambient standards [5]. Hyderabad and Secunderabad, the twin-city of focus of this work, records a mean annual PM_{2.5} concentration of approximately 39 $\mu\text{g}/\text{m}^3$, nearly eight times the WHO guideline of 5 $\mu\text{g}/\text{m}^3$. Still the city is served by fewer than 20 CPCB reference stations for a population exceeding 11 million [3].

Reference-grade monitoring stations achieve high measurement accuracy, but are expensive to deploy and maintain,

significantly limiting their spatial coverage. Low-cost sensors such as the Nova SDS011 nodes considerably reduce hardware cost, but suffer from temperature and humidity-dependent drift that requires continuous calibration to remain reliable [6]. Neither sensor class captures any visual scene information, whereas a human observer can readily judge local air quality by visible haze, sky discoloration, and the density of vehicular exhaust. Camera imagery encodes exactly these cues, and several works have demonstrated that AQI can be estimated directly from traffic or sky images using deep learning methods [7]–[10]. Our own previous work introduced the TRAQID dataset [11] and subsequently the AQIFormer transformer architecture [12] for this task. Despite these advances, three fundamental gaps remain unaddressed across all existing image-based AQI methods.

Existing methods process a single narrow-angle rectilinear image that covers 60° to 90° of azimuth, ignoring the overhead sky, the lateral streetscape, and the rear traffic scene, all of which carry independent atmospheric signals. A 360° equirectangular panorama covers the entire hemisphere simultaneously, providing roughly four times the atmospheric sampling area of a single front-facing view. Second, all prior methods treat the AQI band prediction as a standard multi-class classification problem with categorical cross-entropy loss, which applies equal penalty to adjacent-band errors and extreme-band errors alike, ignoring the natural ordinal structure of AQI categories. Furthermore, there are no publicly available panoramic image datasets for AQI estimation worldwide.

This paper addresses all three gaps. The remainder of this paper is organised as follows: Section III describes the PAN-AQI dataset and its collection campaign, Section IV presents the PANQIFormer architecture, Section V reports the experimental evaluation, and Section VI concludes with limitations and future directions.

II. RELATED WORK

A. Image-Based AQI Estimation

Limited efforts have been devoted to predicting AQI using image-based methods, though research activity has grown significantly in recent years. The physical basis for this approach is that airborne particulate matter scatters and attenuates visible light in proportion to its concentration, producing measurable changes in scene contrast, sky brightness, and colour saturation that correlate with PM levels [15], [16]. Liu

TABLE I
COMPARISON OF IMAGE-BASED AQI DATASETS.

Dataset / Method	Pano	Multi-view	Co-located	Multi-season	# Samples
Mondal et al. [13]	✗	✗	✗	✗	1,818
Liu et al. [7]	✗	✗	✗	NA	6,587
Kow et al. [14]	✗	✗	✗	NA	3,549
Kalajdjieski et al. [8]	✗	✗	✗	✗	1,000
Nilesh et al. [9]	✗	✗	✓	✓	5,048
TRAQID [11]	✗	✓	✓	✓	26,678
PAN-AQI (Ours)	✓	✓	✓	✓	33,982

et al. [7] were among the first to exploit this principle by using brightness and colour features from fixed outdoor scenes to estimate $PM_{2.5}$. Mondal et al. [13] collected smartphone images in Dhaka and trained a compact CNN to predict $PM_{2.5}$, demonstrating that mobile image capture is a viable data collection strategy. Kow et al. [14] gathered images in Kaohsiung, Taiwan, with co-located AQI data and applied deep learning for real-time estimation. Kalajdjieski et al. [8] extended this further by combining InceptionV3 image features with a three-layer MLP over temperature and humidity inputs, showing that incorporating meteorological context measurably improves estimation performance. Zhang et al. [10] proposed AQC-Net, which augments a ResNet-18 backbone with a Spatial and Context Attention block to focus on pollution-relevant regions in the image. For the Indian traffic scenario, Nilesh et al. [9] fused YOLOv5-based vehicle counts with weather features in a Random Forest classifier. Kathalkar et al. [11] introduced TRAQID, the first dual-view front-and-rear traffic image benchmark with co-located multi-season sensor readings, and subsequently developed AQIFormer [12], a transformer-based ensemble with weather-aware attention that achieved 89.96% within-city accuracy on TRAQID and 81.67% on an independent Nagpur test set under few-shot adaptation. Despite this progress, all methods in this line of work operate on narrow rectilinear images and treat AQI band prediction as an unordered multi-class problem. Table I summarises these methods across key dataset and architectural attributes, highlighting the gaps that PAN-AQI and PANQIFormer are designed to fill.

B. Transformer-Based Representation Learning

Dosovitskiy et al. [17] demonstrated that processing images as sequences of fixed-size patches using multi-head self-attention achieves competitive performance with convolutional networks, with the additional benefit of direct long-range interactions between any two patches. Caron et al. [18] showed through the DINO self-distillation objective that ViT patch tokens trained without labels develop emergent semantic properties and transfer well to dense prediction tasks without labelled fine-tuning. DINOv3 [19], trained on approximately 1.7 billion curated images, strengthened these properties further, producing features that generalise to depth estimation, segmentation, and scene understanding with minimal downstream supervision. This makes DINOv3 well-suited for our setting, where the training set size is modest and preserving the pre-

trained feature manifold is important. We therefore freeze the backbone and train only the fusion components and task heads.

Standard categorical cross-entropy is unsuitable for ordered targets because it treats all misclassifications equally regardless of the distance between classes. The CORAL framework [20] addresses this by reformulating K -class ordinal prediction as $K - 1$ cumulative binary tasks sharing a single projection vector, which guarantees rank consistency. The Ranked Probability Score (RPS) provides a distance-aware training signal by penalising predictions proportionally to the squared difference between the predicted and target cumulative distributions over ordered classes. On the sampling side, Buda et al. [21] showed through systematic experiments that combining a class-weighted loss with minority-class oversampling at the batch level consistently outperforms either strategy alone, and Cui et al. [22] reached the same conclusion through their effective-number weighting framework. We apply both in our training design to address the dataset imbalance in PAN-AQI.

C. Contributions

We introduce **PAN-AQI** and **PANQIFormer** for panoramic image-based AQI estimation. Our key contributions are:

- 1) The PAN-AQI dataset comprises 33,982 panoramic 360° images with co-located $PM_{2.5}$, PM_{10} , temperature, humidity and timestamp at every sample, collected over ~ 1000 km across Hyderabad spanning three seasons. To the best of our knowledge, it is the largest open image-based AQI dataset for the Indian subcontinent and the only one in panoramic format with co-located ground-truth PM measurements.
- 2) PANQIFormer is the first image-based AQI framework to integrate 360° panoramic encoding, spatial-zone reasoning, meteorological fusion, and rank-consistent ordinal learning within a unified transformer architecture.
- 3) A comprehensive evaluation demonstrates consistent improvements over prior image-based AQI methods and validates the contributions of panoramic coverage, spatial reasoning, and metadata fusion.

III. THE PAN-AQI DATASET

The PAN-AQI dataset is the primary resource on which PANQIFormer is trained and evaluated. It contains equirectangular panoramic images collected in motion across Hyderabad and Secunderabad road network, each paired with co-located weather readings recorded at the same timestamp. The following subsections describe the capture platform, the collection campaign, the pre-processing pipeline, and the statistical structure of the data.

A. Camera and Sensor Setup

Data collection was carried out using the iHub-Bodhayan [23], an electric SUV-class research platform of IIIT-Hyderabad. The sensing suite is largely identical to our prior work [11], comprising a Nova SDS011 [24] for continuous $PM_{2.5}$ and PM_{10} measurements, an AHT10 sensor [25] for temperature and relative humidity, and an

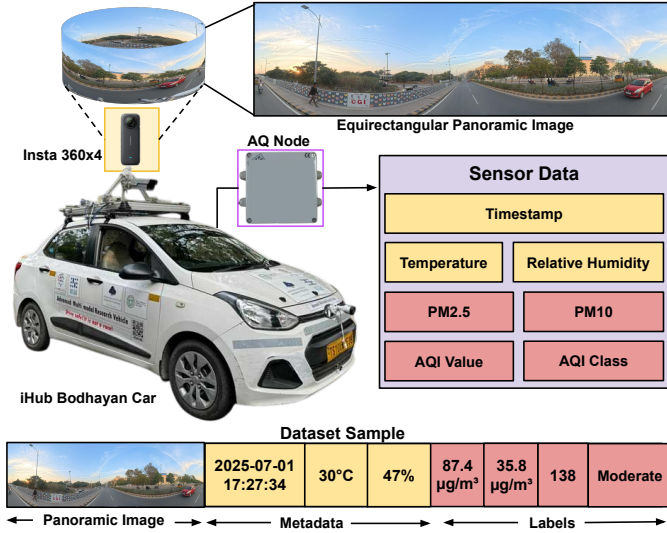


Fig. 1. Data Collection Pipeline

Aeroqual S500 [26] as a periodic reference for sensor validation. The key hardware addition in this work is a roof-mounted Insta360 X4 [27] panoramic camera positioned at 2.1m above ground level, which replaces the dual front-and-rear dashboard cameras of the earlier setup. At this height, the optical axis aligns naturally with the road horizon, keeping vehicle-body reflections outside the effective image area.

The camera records 8K dual-fisheye video and stitches the two hemispheres in-camera into a single equirectangular panorama, providing complete 360° azimuthal coverage at every sample. The Nova SDS011 reports $PM_{2.5}$ and PM_{10} every second over UART shnnand serves as the source of all per-sample ground truth labels. The AHT10 comfortably covers the full range of atmospheric conditions encountered during the campaign, with temperature spanning 25.3°C to 57.1°C and relative humidity from 20.6% to 100.0%. All four streams are time-stamped on a common onboard clock, ensuring precise co-location between imagery and sensor readings at every sample.

B. Dataset Collection Campaign

The collection campaign was conducted over 23 distinct days between December 2024 and August 2025, covering the urban road network across all three Indian seasons. Routes were planned to sample a diverse mix of slow-moving inner-city traffic, free-flow ring-road segments, and signal-controlled intersections, ensuring that the corpus reflects the full range of urban traffic and atmospheric exposure conditions. The seasonal distribution of collected samples is Summer 16,292 (47.9%), Monsoon 10,458 (30.8%), and Winter 7,232 (21.3%). Data was collected exclusively during daytime hours, with Morning contributing 50.5%, Afternoon 27.4%, and Evening 22.1% of the total samples. The absence of nighttime data is acknowledged as a limitation and is discussed in Section VI. Each retained image is paired with an aligned $\langle T, RH, PM_{2.5}, PM_{10} \rangle$ tuple recorded by the cabin

sensors at the same timestamp, from which the cyclic temporal encodings ($\sin h, \cos h, \sin d, \cos d$) used by PANQIFormer (Section IV) are derived.

C. Pre-processing

Per-sample AQI is computed from the co-located PM readings using the same CPCB piecewise-linear sub-index formulation followed in our prior work [3], [11], where the final AQI is taken as the maximum sub-index across $PM_{2.5}$ and PM_{10} . Sensor spikes are removed per season using $1.5 \times IQR$ filtering before AQI computation. After cleaning, the dataset spans $AQI \in [8.4, 434.7]$ with a median of 97.1 and a mean of 122.1 ± 82.3 , while $PM_{2.5}$ and PM_{10} range over $[4.2, 295.1]$ and $[8.4, 308.5] \mu g/m^3$ respectively.

Each panorama undergoes four sequential processing steps. The Insta360 X4 first stitches the dual-fisheye hemispheres into an equirectangular frame at a native resolution of $10,560 \times 5,280$ px, which is then downsampled to a $3,840 \times 1,152$ px archival frame. The bottom 10% of each frame, which captures the vehicle bonnet, is cropped out, yielding a $3,840 \times 1,036$ px archival panorama and incidentally removing the most geometrically distorted polar region of the equirectangular projection. The archival panorama is then resized to 224×672 px for model input, preserving a 1:3 aspect ratio. Finally, per-channel normalisation is applied using the DINOv3 pre-training statistics [19].

D. Dataset Description and Statistical Analysis

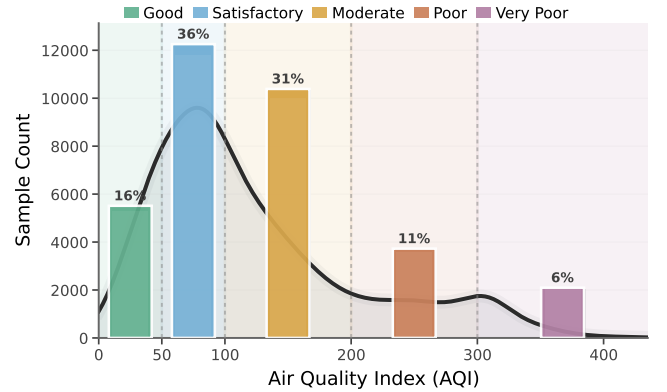


Fig. 2. Dataset Composition

Following the CPCB standard [3], AQI values are assigned to five bands. The Severe class ($AQI > 400$) contains only 42 samples (0.12%), well below the threshold at which a class is statistically defensible [21], so we merge it into Very Poor, resulting in five classes with breakpoints at $\{50, 100, 200, 300\}$. The resulting five-class distribution exhibits moderate imbalance, with a $5.85\times$ ratio between the largest and smallest classes. The seasonal variation is pronounced: the per-season mean AQI is 220.93 ± 81.20 for Winter, 118.99 ± 58.89 for Summer, and 58.55 ± 34.79 for Monsoon. Winter loads the rare classes heavily, with 57.34% of its samples falling in the Poor or VP+ bands and contributing approximately 79% of the dataset's VP+ volume, while 87.52% of Monsoon samples fall in the Good or Satisfactory bands. Morning carries

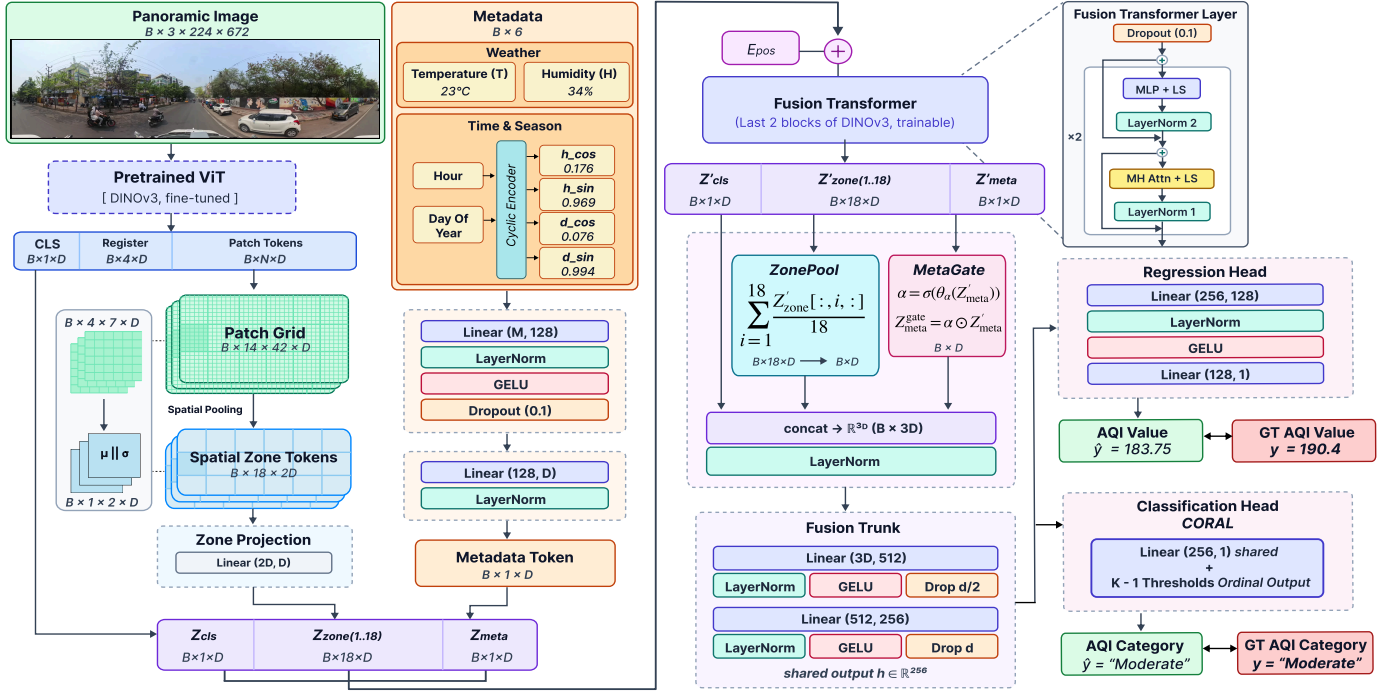


Fig. 3. Architecture Diagram

nearly all the VP+ volume across the day, while Afternoon is the most class-uniform period. These seasonal and diurnal structures, visualised in Figure 2, directly motivate the fusion of meteorological and temporal metadata alongside image features in PANQIFormer.

IV. METHODOLOGY

We propose a multi-task deep learning framework for joint AQI value regression and classification from panoramic traffic images and meteorological metadata. As illustrated in Figure 3, the architecture comprises three stages: visual encoding with spatial zone pooling, multi-modal fusion via a fusion transformer, and prediction heads from a shared latent representation.

A. Visual Encoding and Spatial Zone Representation

Input panoramic images of size $3 \times 224 \times 672$ are processed using a DINOv3 ViT-B/16 backbone, producing patch embeddings of dimension $D = 768$, together with a global [CLS] token and four register tokens. The register tokens are discarded since they function as non-spatial computation tokens rather than representations of image regions [28].

Unlike perspective images, pollution cues in panoramic scenes may appear across the full azimuth, making spatial localisation important. Simple global pooling over the patch tokens would suppress localised environmental variability, particularly across sky, mid-field, and ground regions, where atmospheric cues may exhibit distinct visual characteristics. To preserve coarse spatial structure while maintaining a compact representation, the patch embeddings are partitioned into a fixed 6×3 spatial layout, where each zone aggregates a local group of neighbouring patches.

For each spatial zone, mean and standard deviation statistics are computed across the constituent patch embeddings. The mean captures dominant regional semantics, while the standard deviation encodes local feature variability arising from heterogeneous atmospheric structure, illumination, and scene texture. Together, these statistics provide a compact representation of both regional appearance and intra-zone heterogeneity. The resulting zone descriptors are projected into the shared latent space to form 18 spatial tokens, which are retained alongside the global [CLS] representation for subsequent multimodal fusion.

B. Metadata Integration and Multi-Modal Fusion

Similar visual scenes may arise under different atmospheric conditions, motivating the incorporation of environmental metadata. PANQIFormer therefore incorporates meteorological and temporal metadata alongside visual features. Temperature and relative humidity are provided directly as scalar inputs, while season and time of day are encoded using standard sine-cosine cyclic transformations of the fractional day-of-year and hour-of-day, producing four continuous features that preserve diurnal and seasonal periodicity without discontinuities at cycle boundaries. The resulting six-dimensional metadata vector is projected into a token through two learned linear layers with LayerNorm, forming a single metadata token of dimension D .

Metadata is represented as a token to enable joint attention-based fusion with visual representations. Learned positional embeddings are added to the [CLS], spatial-zone, and metadata tokens before joint processing by the Fusion Transformer, implemented using the final two fully trainable blocks of

DINOv3. It allows the fusion module to adapt to the joint visual-metadata token distribution.

The transformed tokens of the [CLS], zone, and metadata streams are retained as separate representations. The [CLS] token preserves global scene semantics, zone tokens are average-pooled into a descriptor summarising spatial context, and the metadata token is passed through a gating module that adaptively re-weights channels of the fused meteorological embedding. Separate global, spatial, and metadata streams preserve complementary information prior to final fusion. The resulting representations are concatenated, layer-normalised, and projected through a two-layer Fusion Trunk to produce a shared latent representation h .

C. Multi-Task Prediction and Training

The shared latent representation h is jointly optimised for continuous AQI regression and ordinal category prediction, encouraging sensitivity to both fine-grained pollution variation and ordered air-quality structure. The regression head predicts a continuous AQI value. Regression targets are log-transformed to mitigate the long-tailed target distribution and optimised using Smooth L1 loss.

The classification branch follows the CORAL framework [20], which models ordinal AQI categories using rank-consistent cumulative logits. These cumulative predictions are converted into class probabilities and supervised using the Ranked Probability Score (RPS), which penalises errors according to ordinal severity. Thus, CORAL defines the ordinal parameterisation, while RPS serves as the training objective. To mitigate class imbalance, inverse-frequency weighting is applied to the marginal class probabilities within the RPS loss.

The overall training objective combines ordinal classification and regression losses:

$$\mathcal{L} = \mathcal{L}_{\text{RPS}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}, \quad \lambda_{\text{reg}} = 0.3 \quad (1)$$

Optimization uses AdamW with staged unfreezing. Training begins with only the prediction heads active, while the pretrained DINOv3 backbone remains frozen. The Fusion Transformer is unfrozen at epoch 5 to adapt the pretrained visual features to the joint visual-metadata token distribution, followed by unfreezing of the final DINOv3 backbone block at epoch 10 for task-specific refinement.

V. RESULTS AND DISCUSSION

This section evaluates PANQIFormer against prior image-based AQI estimation methods and analyses the contributions of architectural components through ablation studies.

A. Comparison with Prior Methods

Table II compares our method against representative image-based AQI estimation approaches. Our method outperforms the strongest prior method, AQIFormer [12], by 5.2% in accuracy and 6.1% in macro F1-score, while reducing the MAE for AQI by 11.14. It achieves an overall best performance across all evaluation metrics, reaching 88.9% accuracy, 87.3% macro F1-score, and an MAE of 10.45.

TABLE II
QUANTITATIVE COMPARISON OF IMAGE-BASED AQI ESTIMATION
METHODS ON PAN-AQI

Method	Acc $\uparrow$	F1 $\uparrow$	MAE $\downarrow$
Mondal et al. [13]	64.0	62.0	43.98
Kalajdjieski et al. [8]	71.1	68.4	32.8
Nilesh et al. [9]	80.0	76.0	27.61
AQC-Net [10]	72.0	61.0	28.64
Kathalkar et al. [12]	83.7	81.2	21.59
PANQIFormer (ours)	88.9	87.3	10.45

The consistency of these gains across both classification and regression metrics suggests that PANQIFormer captures not only AQI category boundaries but also the underlying continuum of pollution severity. It achieves 97.4% agreement between the two heads, indicating highly consistent prediction across the tasks. Notably, the gains persist over recent weather-aware and transformer-based approaches, showing that the gains arise from the proposed panoramic representation and multimodal reasoning framework rather than stronger visual backbones alone. These results support the central hypothesis of this work: preserving full panoramic context provides substantially richer environmental information than narrow field-of-view imagery for AQI estimation.

B. Ablation Study

Table III evaluates the contribution of panoramic coverage and architectural components. The performance improves monotonically as field of view expands from single-view to panoramic imagery, with single, dual, and four-view variants consistently underperforming the full panorama. While adding views progressively reduces the performance gap from 4.4% to 1.9%, a consistent deficit remains, indicating that multi-view aggregation cannot fully reconstruct the continuous environmental context preserved within a unified 360° panorama. These findings validate the core premise of PAN-AQI and demonstrate that distributed atmospheric cues benefit from being observed within a unified panoramic representation.

Among architectural components, the DINOv3 backbone and spatial-zone representation contribute the largest gains. Replacing DINOv3 with ResNet-50 reduces accuracy by 6.3%, highlighting the importance of large-scale self-supervised representations and their capacity to capture enriched features. Similarly, removing zone encoding reduces accuracy by 4.1% and increases MAE by 5.65. It shows that localised spatial structure remains highly informative despite AQI being a globally defined target. Experimenting with mean-only pooling degrades performance by 1.2% accuracy, showing that intra-zone feature variability provides complementary information beyond dominant regional content.

Metadata contributes smaller, but consistent, gains relative to the visual components. Removing metadata reduces accuracy by 1.1% and increases MAE by 2.35, indicating that atmospheric measurements provide useful context not directly observable from imagery alone. The limited degradation suggests that panoramic imagery captures much of AQI-related signals itself. Likewise, replacing the Fusion Transformer with

TABLE III
ABLATION STUDY OF PANORAMIC COVERAGE AND
ARCHITECTURAL COMPONENTS

Category	Configuration	Acc $\uparrow$	F1 $\uparrow$	MAE $\downarrow$	Δ Acc
Panoramic Coverage	Single-view (front)	84.5	82.4	13.07	-4.4
	Dual-view (front+rear)	85.9	83.8	12.30	-3.0
	Four-view	87.0	85.3	11.92	-1.9
Architectural Variants	ResNet-50 Backbone	82.6	80.2	18.70	-6.3
	Without Metadata Encoding	87.8	85.6	12.80	-1.1
	Without Zone Encoding	84.8	83.0	16.10	-4.1
	Mean-only zone pooling	87.7	86.2	11.82	-1.2
	Concatenation Fusion	84.0	81.5	17.20	-4.9
	MLP Fusion	83.4	80.7	17.83	-5.5
PANQIFormer		88.9	87.3	10.45	—

concatenation or MLP-based fusion reduces accuracy by 4.9% and 5.5% while increasing MAE by 6.75 and 7.38, respectively. This suggests that effective AQI estimation requires modelling cross-modal interactions rather than simple feature aggregation.

Taken together, the ablation study indicates that the gains of PANQIFormer arise from improvements at multiple levels. Panoramic coverage preserves environmental context that is inaccessible to narrow field-of-view imagery, spatial-zone reasoning retains localised atmospheric structure within the panorama, and attention-based fusion enables interactions between visual observations and meteorological conditions. These components contribute consistently across the ablations and collectively explain the performance improvements observed in Table II, validating the principal design choices underlying our method.

VI. CONCLUSION

We presented PAN-AQI, the first large-scale panoramic dataset for image-based AQI estimation in India, and PAN-QIFormer, a multimodal transformer framework that jointly models panoramic visual context and environmental metadata. The experiments demonstrate consistent improvements over prior image-based AQI methods and show that preserving full 360° scene context, explicit spatial reasoning, and attention-based multimodal fusion each contribute substantially to performance. Collectively, these findings suggest that panoramic sensing provides a more effective representation for vision-based environmental monitoring than conventional perspective imagery. We hope PAN-AQI serves as a benchmark for future research in scalable, image-driven urban air-quality assessment.

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